

ideaspark

ESP32 2.8 inch LCD Board

Pin Reference · Touch and Non-Touch Versions · R1.0

English version — Click here to access the file: <https://github.com/GJKJ/ESP3228LCD>

Versión en español — Haga clic aquí para acceder al archivo: <https://github.com/GJKJ/ESP3228LCD>

النسخة العربية — انقر هنا للوصول إلى الملف: <https://github.com/GJKJ/ESP3228LCD>

Versão em português — Clique aqui para acessar o arquivo: <https://github.com/GJKJ/ESP3228LCD>

Русская версия — Нажмите здесь, чтобы перейти к файлу: <https://github.com/GJKJ/ESP3228LCD>

日本語版 — ファイルはこちらをクリックしてください: <https://github.com/GJKJ/ESP3228LCD>

Version française — Cliquez ici pour accéder au fichier : <https://github.com/GJKJ/ESP3228LCD>

Deutsche Version — Klicken Sie hier, um auf die Datei zuzugreifen: <https://github.com/GJKJ/ESP3228LCD>

한국어 버전 — 파일을 보려면 여기를 클릭하세요: <https://github.com/GJKJ/ESP3228LCD>

This board is offered in two versions that share the same PCB, the same header layout and the same display. The touch version fits an XPT2046 resistive touch controller; the non-touch version does not. Everything in this document that differs between them is marked.

The header layout, pin names and pin functions are identical to the ESP32 DEVKIT V1, so pin numbers taken from existing ESP32 tutorials apply directly. The difference is that some of those GPIO pins are already wired to the display, the touch controller and the microSD slot. This document lists exactly which ones, what is left for you, and what to watch out for.

1. The two versions at a glance

	Touch version	Non-touch version
Display	2.8 in, 240 x 320, ST7789	2.8 in, 240 x 320, ST7789
Touch controller	XPT2046, resistive	Not fitted
microSD slot	Yes, on the shared SPI bus	Yes, on the shared SPI bus
GPIO used by the board	10	8
Free GPIO (input and output)	9	11
Free analog inputs (input only)	4	4
Devices on the SPI bus	3	2

The two pins that differ are **GPIO14** and **GPIO27**. On the touch version they carry the touch chip select and the touch interrupt. On the non-touch version they are free for your own use.

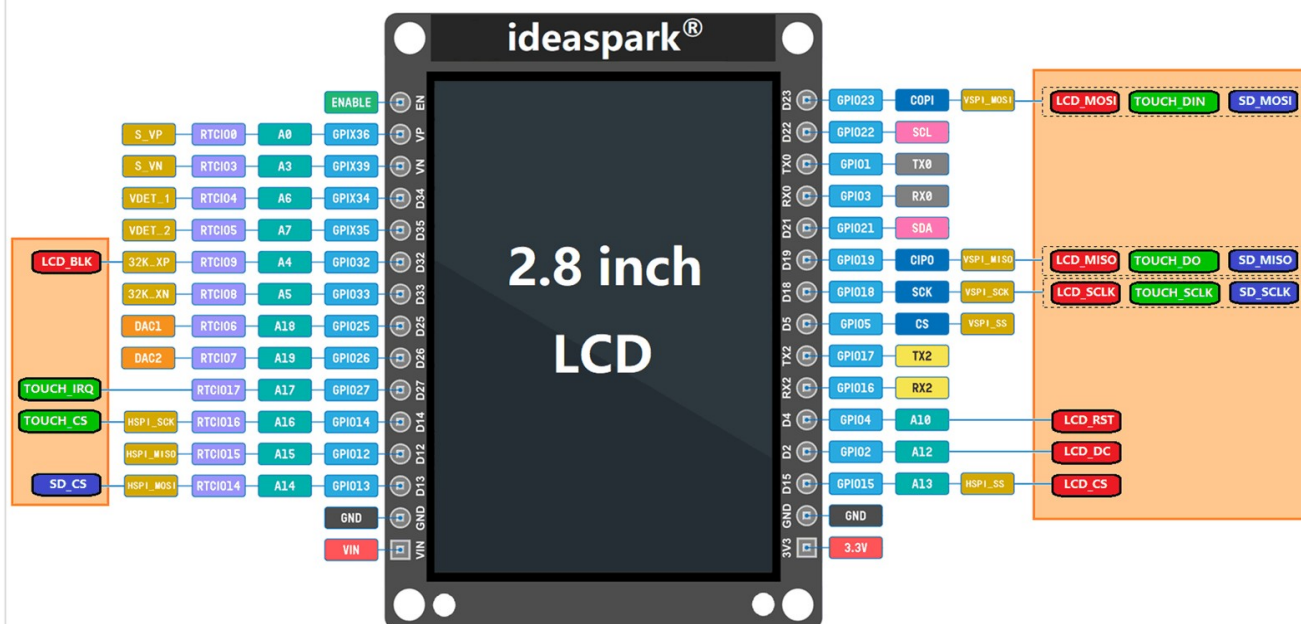
2. Pinout diagrams

Each version has its own diagram. The touch version shows three peripheral groups on the shared SPI bus; the non-touch version shows two, and GPIO14 and GPIO27 carry no assignment. View on screen and zoom in for the small labels.

ideaspark® ESP32 2.8 inch Touch LCD SD Card Board

PINOUT

- LCD Screen to ESP32 Connections
- Touch to ESP32 Connections
- SD Card to ESP32 Connections

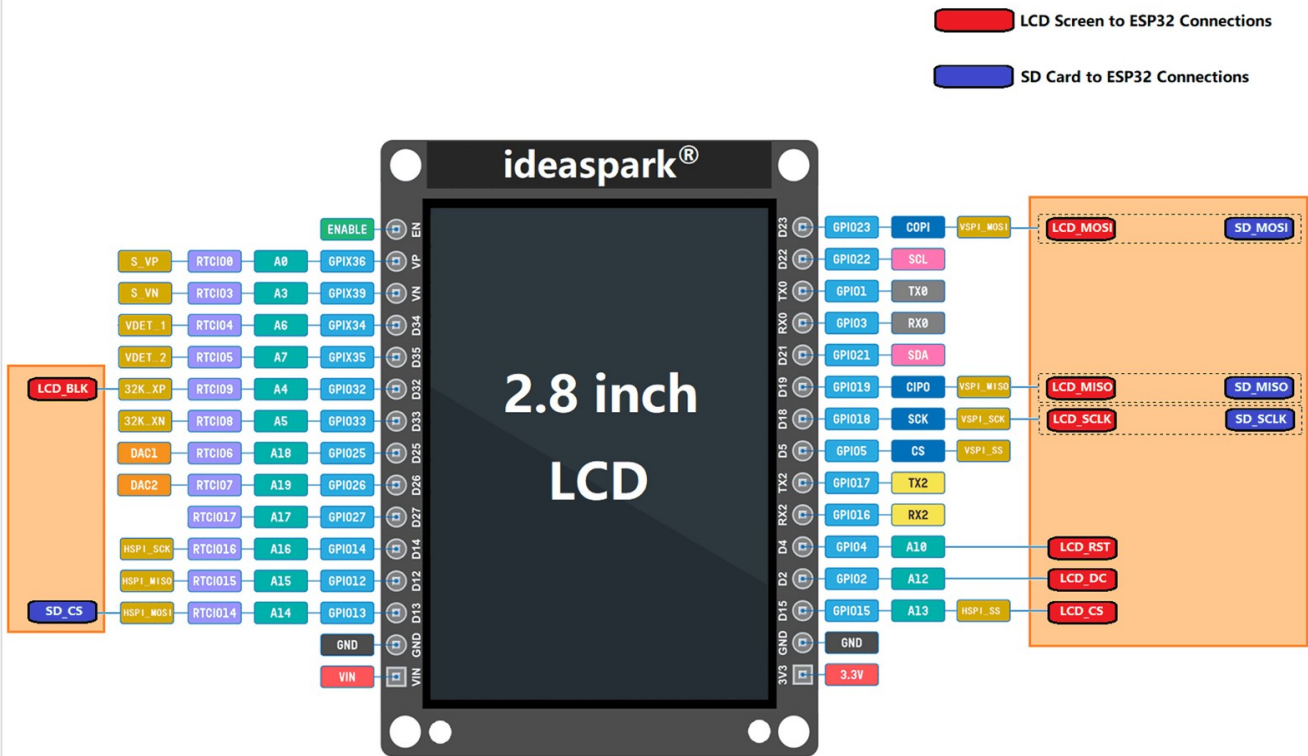


- GPIO pins 34, 35, 36 and 39 are input only. GPIO pins are Input and Output
 - TX0 and RX0 (Serial0) are used for serial programming.
 - TX2 and RX2 can be accessed as Serial2.
 - DAC: Digital-to-Analog Converter. ADC INPUTS: Analog-to-Digital Converter
 - Default SPI is VSP1. Both VSP1 and HSP1 pins can be set to any GPIO pins.
 - All GPIO pins support PWM and interrupts.
 - Built-in LED is connected to GPIO2.
 - Some GPIO pins are used for interfacing flash memory and thus are not shown.
- IDEASARK ESP32 2.8" LCD Board pins name and function are the same as "ESP32 DEVKIT V1"

PHYSICAL PIN	DAC OUTPUTS
CONTROL PINS	I2C PINS
GPIO PINS	SPI PINS
POSITIVE SUPPLY	UART PINS
GROUND SUPPLY	EXCLUDED PINS
ADC INPUTS	Other Related Functions

Touch version — GPIO14 and GPIO27 carry the touch chip select and touch interrupt

ideaspark® ESP32 2.8 inch LCD SD Card Board **PINOUT**



- GPIO pins 34, 35, 36 and 39 are input only. GPIO pins are Input and Output
- TX0 and RX0 (Serial0) are used for serial programming.
- TX2 and RX2 can be accessed as Serial2.
- DAC: Digital-to-Analog Converter. ADC INPUTS: Analog-to-Digital Converter
- Default SPI is VSP1. Both VSP1 and HSP1 pins can be set to any GPIO pins.
- All GPIO pins support PWM and interrupts.
- Built-in LED is connected to GPIO2.
- Some GPIO pins are used for interfacing flash memory and thus are not shown.
- IDEASPARK ESP32 2.8" LCD Board pins name and function are the same as "ESP32 DEVKIT V1"

PHYSICAL PIN	DAC OUTPUTS
CONTROL PINS	I2C PINS
GPIO PINS	SPI PINS
POSITIVE SUPPLY	UART PINS
GROUND SUPPLY	EXCLUDED PINS
ADC INPUTS	Other Related Functions

Non-touch version — GPIO14 and GPIO27 are free for your own use

What changes between the two diagrams

The green Touch group disappears from the legend and from the shared SPI bus.

TOUCH_CS on GPIO14 and TOUCH_IRQ on GPIO27 are removed, leaving both pins free.

TOUCH_DIN, TOUCH_DO and TOUCH_SCLK are removed from GPIO23, GPIO19 and GPIO18. Those three pins stay in use by the display and the microSD slot on both versions.

Everything else is identical: the same PCB, the same header layout, the same display, and the same microSD slot.

3. Header layout

Two rows of 15 pins at 0.1 inch (2.54 mm) pitch.

Header pin	Function on this board	Touch version	Non-touch version
EN	Reset input. Pull LOW to reset the ESP32	Power	Power
GPI036	Analog input, input only	Free (in)	Free (in)
GPI039	Analog input, input only	Free (in)	Free (in)
GPI034	Analog input, input only	Free (in)	Free (in)
GPI035	Analog input, input only	Free (in)	Free (in)
GPI032	LCD backlight (LCD_BLK)	Used	Used
GPI033	General purpose, ADC1 channel	Free	Free
GPI025	General purpose, DAC1 output	Free	Free
GPI026	General purpose, DAC2 output	Free	Free
GPI027	Touch interrupt (TOUCH_IRQ)	Used	Free
GPI014	Touch chip select (TOUCH_CS)	Used	Free
GPI012	General purpose. Strapping pin, see section 6	Free	Free
GPI013	microSD chip select (SD_CS)	Used	Used
GND	Ground	Power	Power
VIN	5 V supply input	Power	Power
GPI023	Shared SPI data out (MOSI)	Used	Used
GPI022	General purpose, default I2C clock (SCL)	Free	Free
GPI01	TX0, USB programming serial port	Programming	Programming
GPI03	RX0, USB programming serial port	Programming	Programming
GPI021	General purpose, default I2C data (SDA)	Free	Free
GPI019	Shared SPI data in (MISO)	Used	Used
GPI018	Shared SPI clock (SCLK)	Used	Used
GPI05	General purpose. Strapping pin, see section 6	Free	Free
GPI017	General purpose, TX2 of the second UART	Free	Free
GPI016	General purpose, RX2 of the second UART	Free	Free
GPI04	LCD reset (LCD_RST)	Used	Used
GPI02	LCD data / command select (LCD_DC)	Used	Used
GPI015	LCD chip select (LCD_CS)	Used	Used
GND	Ground	Power	Power
3V3	3.3 V rail	Power	Power

Programming pins. GPIO1 (TX0) and GPIO3 (RX0) carry the USB programming serial port. They are broken out and can be repurposed, but doing so interferes with uploading sketches and with the serial monitor. They are not counted as free pins.

4. What is free for you

	Pins									
Touch version	5	12	16	17	21	22	25	26	33	
Non-touch version	5	12	14	16	17	21	22	25	26	27 33
Analog in, both versions	34	35	36	39						

Pins 34, 35, 36 and 39 are input only. They cannot drive an output and have no internal pull-up or pull-down, so a button or an open-collector sensor on one of them needs an external resistor.

5. Peripherals still available

I2C

Both default pins are free on either version, so `Wire.begin()` works with no arguments.

Signal	GPIO
SDA	21
SCL	22

Second hardware UART

Both default pins are free on either version. Available in Arduino as `Serial2`.

Signal	GPIO
RX2	16
TX2	17

DAC

Both 8-bit digital to analog outputs are free on either version.

Channel	GPIO
DAC1	25
DAC2	26

Analog input

This is the one area that needs planning. The ESP32 has two ADC blocks, and ADC2 cannot be read while Wi-Fi is active. Most of the free analog-capable pins on this board sit on ADC2.

GPIO	ADC block	Readable with Wi-Fi on?	Touch version	Non-touch version
33	ADC1	Yes	Free	Free
34	ADC1	Yes	Free (input only)	Free (input only)

GPIO	ADC block	Readable with Wi-Fi on?	Touch version	Non-touch version
35	ADC1	Yes	Free (input only)	Free (input only)
36	ADC1	Yes	Free (input only)	Free (input only)
39	ADC1	Yes	Free (input only)	Free (input only)
12	ADC2	No	Free	Free
25	ADC2	No	Free	Free
26	ADC2	No	Free	Free
14	ADC2	No	Used by touch	Free
27	ADC2	No	Used by touch	Free
32	ADC1	Yes	Used by backlight	Used by backlight
13	ADC2	No	Used by microSD	Used by microSD
15	ADC2	No	Used by display	Used by display
4	ADC2	No	Used by display	Used by display
2	ADC2	No	Used by display	Used by display

If your project reads analog sensors while connected to Wi-Fi

Use GPIO 33, 34, 35, 36 and 39. That is five channels on both versions, one of which (GPIO33) can also drive outputs.

Freeing GPIO14 and GPIO27 on the non-touch version does not add Wi-Fi-safe analog inputs, because both sit on ADC2. It does add two more general purpose pins.

Capacitive touch channels

Separate from the XPT2046 resistive panel, the ESP32 chip itself has capacitive touch channels. Most land on pins this board already uses.

Channel	GPIO	Touch version	Non-touch version
T5	12	Available	Available
T8	33	Available	Available
T6	14	Not available	Available
T7	27	Not available	Available

Channels T0, T2, T3, T4 and T9 fall on GPIO 4, 2, 15, 13 and 32, which are used by the display and the microSD slot on both versions.

A second SPI device

The bus is already in use, but the ESP32 routes SPI through a GPIO matrix, so a second SPI interface can be assigned to any free pins. Alternatively, add your device to the existing bus with its own chip select on a free pin.

PWM and interrupts

Every free output-capable pin supports PWM. Every free pin, including the input-only ones, supports external interrupts.

6. Cautions

GPIO12 — read this before wiring anything to it

GPIO12 is a strapping pin (MTDI). It selects the flash supply voltage at power-up. If it is held HIGH when the board boots, the ESP32 will try to run its 3.3 V flash at 1.8 V and will fail to start. The board will appear dead.

Anything connected to GPIO12 must leave it low or floating at power-up. A sensor module with a pull-up resistor on that line will stop the board booting. This applies to both versions.

GPIO14 — brief pulse at boot

GPIO14 outputs a short PWM burst during boot.

On the touch version this pin is the touch chip select and the pulse is harmless.

On the non-touch version the pin is yours, and a relay or a servo attached to it will twitch once at power-up. Add a pull-down, or choose a different pin for anything that must not move on start-up.

GPIO5 — strapping pin

GPIO5 is internally pulled up and is sampled at boot. It is safe for general input and output, but avoid connecting anything sensitive to a brief high level during the first moments after power-up. This applies to both versions.

GPIO2 and GPIO15 during upload

GPIO2 (LCD_DC) must be low or floating to enter the serial bootloader, and GPIO15 (LCD_CS) silences the ROM boot messages when held low.

Both are driven by the display library only after `init()` runs, so normal uploads are unaffected. If you drive either pin externally, uploads may fail or the boot log may disappear.

GPIO16 and GPIO17

These two pins are free on this board because the module carries no PSRAM. On ESP32 modules that do include PSRAM, these pins are consumed by it.

7. Working with the shared SPI bus

The display and the microSD card share one SPI bus. On the touch version the touch controller shares it too. Three rules keep them from interfering with each other.

1. Deassert the chip selects you are not using. Set the idle CS pins HIGH before initialising another peripheral. A floating chip select will corrupt data on the shared MISO line. On the touch version that means LCD_CS (15), TOUCH_CS (14) and SD_CS (13); on the non-touch version, LCD_CS (15) and SD_CS (13). This is the most common cause of a microSD card failing to initialise.
2. Let the libraries manage the clock rate. Each peripheral opens an SPI transaction at its own frequency: the display runs fast, the touch controller runs slow, and the card sits between them. As long as every access happens inside a transaction, the bus arbitrates itself.

3. Turn the backlight on before you look for a fault. GPIO32 must be driven HIGH. A board with a working display and a floating backlight pin looks identical to a dead board.

8. Quick reference for Arduino

Touch version

```
#define LCD_MOSI    23    // shared
#define LCD_MISO    19    // shared
#define LCD_SCLK    18    // shared
#define LCD_CS      15
#define LCD_DC       2
#define LCD_RST      4
#define LCD_BLK     32

#define TOUCH_CS    14
#define TOUCH_IRQ   27

#define SD_CS       13

// Free for your own use
// Full GPIO : 5, 12, 16, 17, 21, 22, 25, 26, 33
// Input only: 34, 35, 36, 39
```

Non-touch version

```
#define LCD_MOSI    23    // shared
#define LCD_MISO    19    // shared
#define LCD_SCLK    18    // shared
#define LCD_CS      15
#define LCD_DC       2
#define LCD_RST      4
#define LCD_BLK     32

#define SD_CS       13

// Free for your own use
// Full GPIO : 5, 12, 14, 16, 17, 21, 22, 25, 26, 27, 33
// Input only: 34, 35, 36, 39
```

Display resolution is 240 x 320 pixels with an ST7789 controller. On the touch version the touch controller is an XPT2046 resistive type. The microSD slot is driven over SPI and expects a FAT32 formatted card.

This document describes the ideaspark ESP32 2.8 inch LCD Board in its touch and non-touch versions. Pin function details for the ESP32 itself are drawn from the Espressif ESP32 datasheet and technical reference manual.